

Inference for Linear Regression

IMS Chapter 24

Where we are

408 vs. this class

You have a separate course on *fitting* and *diagnosing* regression models. Here we ask the question we have asked all semester:

If I had drawn a different sample, how different would my answer be?

408 vs. this class

You have a separate course on *fitting* and *diagnosing* regression models. Here we ask the question we have asked all semester:

If I had drawn a different sample, how different would my answer be?

The estimate b_1 (slope) is a **statistic**, just like $\hat{p}$ and $\bar{x}$. So it has a **sampling distribution**, and everything we know carries over:

- ▶ hypothesis test (randomization or math model)
- ▶ confidence interval (bootstrap or math model)
- ▶ conditions that tell us when to trust each

The same recipe, a new parameter

Parameter	Estimate	Null value
p	$\hat{p}$	p_0
μ	$\bar{x}$	μ_0
β_1 (slope)	b_1	0

The same recipe, a new parameter

Parameter	Estimate	Null value
p	$\hat{p}$	p_0
μ	$\bar{x}$	μ_0
β_1 (slope)	b_1	0

Test statistic for a math model:

$$T = \frac{\text{estimate} - \text{null value}}{SE}$$

The model and its estimates

Population model vs. fitted line

Population (unknown): $y = \beta_0 + \beta_1 x + \varepsilon$

Sample (what we compute): $\hat{y} = b_0 + b_1 x$

- ▶ b_0, b_1 are **point estimates** of β_0, β_1 .
- ▶ ε is the part of y the line does not explain.

Population model vs. fitted line

Population (unknown): $y = \beta_0 + \beta_1 x + \varepsilon$

Sample (what we compute): $\hat{y} = b_0 + b_1 x$

- ▶ b_0, b_1 are **point estimates** of β_0, β_1 .
- ▶ ε is the part of y the line does not explain.

The most common question: **is there any linear relationship at all?**

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

A hypothetical: sandwich store revenue

Suppose we *knew* the truth for all stores:

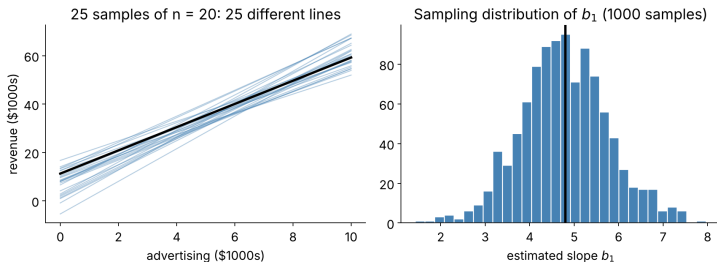
$$\text{expected revenue} = 11.23 + 4.8 \times \text{advertising}$$

A hypothetical: sandwich store revenue

Suppose we *knew* the truth for all stores:

$$\text{expected revenue} = 11.23 + 4.8 \times \text{advertising}$$

Now take random samples of $n = 20$ stores and fit a line each time.

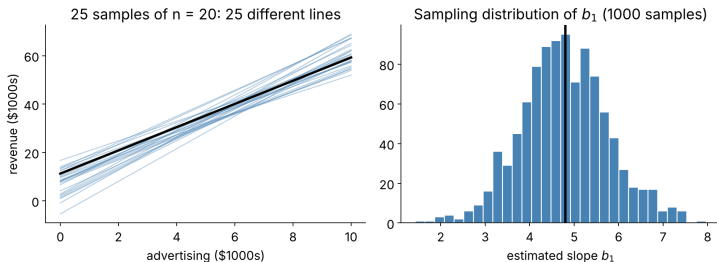


A hypothetical: sandwich store revenue

Suppose we *knew* the truth for all stores:

$$\text{expected revenue} = 11.23 + 4.8 \times \text{advertising}$$

Now take random samples of $n = 20$ stores and fit a line each time.



Slopes vary from sample to sample, centered near the true 4.8. **That spread is what the standard error measures.**

Randomization test for the slope

births14: weight vs. weeks

100 births sampled from births14. Does gestation length help predict birth weight?

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

births14: weight vs. weeks

100 births sampled from births14. Does gestation length help predict birth weight?

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

Observed slope: $b_1 = 0.335$ lb per week.

How could we simulate a world where $\beta_1 = 0$?

births14: weight vs. weeks

100 births sampled from births14. Does gestation length help predict birth weight?

$$H_0 : \beta_1 = 0 \quad H_A : \beta_1 \neq 0$$

Observed slope: $b_1 = 0.335$ lb per week.

How could we simulate a world where $\beta_1 = 0$?

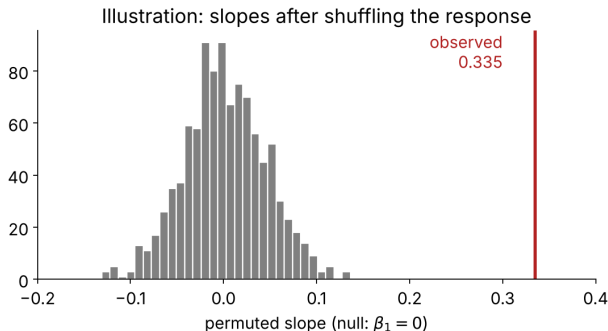
Shuffle the response. If weeks and weight are unrelated, then any weight could have gone with any number of weeks.

The permutation procedure

1. Keep the weeks column fixed.
2. Shuffle the weight column, breaking any link between the two.
3. Fit the line, record the slope.
4. Repeat many times to build the **null distribution**.
5. The p-value is the fraction of permuted slopes at least as extreme as the observed one.

The permutation procedure

1. Keep the weeks column fixed.
2. Shuffle the weight column, breaking any link between the two.
3. Fit the line, record the slope.
4. Repeat many times to build the **null distribution**.
5. The p-value is the fraction of permuted slopes at least as extreme as the observed one.



Bootstrap interval for the slope

Bootstrap: resample *rows*

A bootstrap sample draws **whole observations** (x_i, y_i) with replacement, so each point keeps its partner.

1. Resample n rows with replacement.
2. Fit the line, record b_1^* .
3. Repeat 1000+ times.

Bootstrap: resample *rows*

A bootstrap sample draws **whole observations** (x_i, y_i) with replacement, so each point keeps its partner.

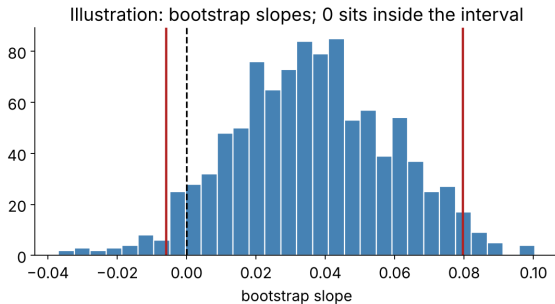
1. Resample n rows with replacement.
2. Fit the line, record b_1^* .
3. Repeat 1000+ times.

Two ways to build the interval:

- ▶ **Percentile:** middle 95% of bootstrap slopes.
- ▶ **SE method:** $b_1 \pm 2 \times SE_{boot}$ (or t^* instead of 2).

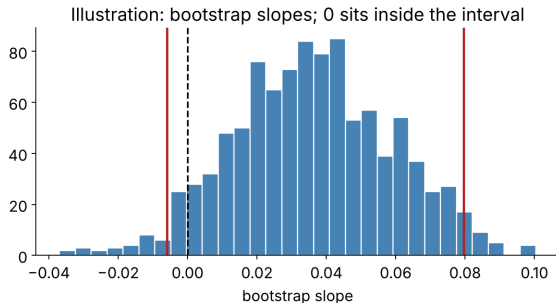
Example: weight vs. mother's age

`weight ~ mage` in the same 100 births. Observed slope 0.036.



Example: weight vs. mother's age

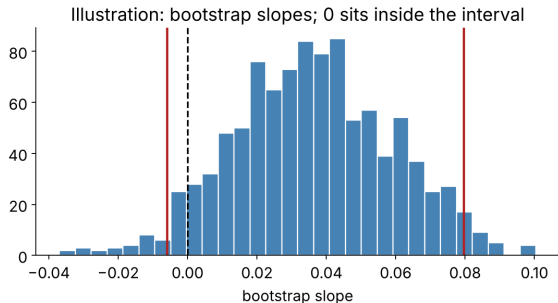
weight $\sim$ mage in the same 100 births. Observed slope 0.036.



- ▶ Bootstrap percentile 95% CI: $(-0.01, 0.081)$
- ▶ Bootstrap SE ≈ 0.0225 , so $0.036 \pm 2(0.0225)$ gives about $(-0.009, 0.081)$

Example: weight vs. mother's age

weight $\sim$ mage in the same 100 births. Observed slope 0.036.



- ▶ Bootstrap percentile 95% CI: $(-0.01, 0.081)$
- ▶ Bootstrap SE ≈ 0.0225 , so $0.036 \pm 2(0.0225)$ gives about $(-0.009, 0.081)$

The interval contains 0. There is not convincing evidence that mother's age is linearly related to birth weight in these data.

Intervals and tests agree

The 95% CI contains 0. What does the two-sided test of $\beta_1 = 0$ say at $\alpha = 0.05$?

Intervals and tests agree

The 95% CI contains 0. What does the two-sided test of $\beta_1 = 0$ say at $\alpha = 0.05$?

We **fail to reject**. Same logic as for means and proportions: a value inside the 95% interval would not be rejected at the 5% level.

Math model: the t-test for the slope

Reading regression output

weight ~ weeks:

term	estimate	std.error	statistic	p.value
(Intercept)	-5.72	1.61	-3.54	0.0006
weeks	0.34	0.04	8.07	< 0.0001

Reading regression output

weight ~ weeks:

term	estimate	std.error	statistic	p.value
(Intercept)	-5.72	1.61	-3.54	0.0006
weeks	0.34	0.04	8.07	< 0.0001

- ▶ $\text{statistic} = \frac{\text{estimate} - 0}{\text{std.error}}$, e.g. $\frac{0.335}{0.0415} \approx 8.07$
- ▶ p.value is **two-sided**, for $H_0 : \beta = 0$
- ▶ Degrees of freedom: $n - 2$ (we estimated two parameters)

Reading regression output

weight ~ weeks:

term	estimate	std.error	statistic	p.value
(Intercept)	-5.72	1.61	-3.54	0.0006
weeks	0.34	0.04	8.07	< 0.0001

- ▶ $\text{statistic} = \frac{\text{estimate} - 0}{\text{std.error}}$, e.g. $\frac{0.335}{0.0415} \approx 8.07$
- ▶ p.value is **two-sided**, for $H_0 : \beta = 0$
- ▶ Degrees of freedom: $n - 2$ (we estimated two parameters)

The intercept row tests $\beta_0 = 0$. Usually that is *not* the interesting question.

Another example: weight vs. mother's age

term	estimate	std.error	statistic	p.value
⋮				
mage	0.036	0.024	1.50	0.1362

Another example: weight vs. mother's age

term	estimate	std.error	statistic	p.value
⋮				
mage	0.036	0.024	1.50	0.1362

$T = 0.036/0.024 = 1.50$ with 98 df gives $p = 0.136$. **Fail to reject.** This matches the bootstrap interval that contained 0.

Confidence interval for a slope

$$b_1 \pm t_{df}^* \times SE$$

with $df = n - 2$.

Confidence interval for a slope

$$b_1 \pm t_{df}^* \times SE$$

with $df = n - 2$.

Example: Elmhurst College. Gift aid vs. family income (in \$1000s), $n = 50$:

$$\text{estimate} = -0.0431, \quad SE = 0.0108, \quad t_{48}^* = 2.01$$

$$-0.0431 \pm 2.01(0.0108) = (-0.0648, -0.0214)$$

Confidence interval for a slope

$$b_1 \pm t_{df}^* \times SE$$

with $df = n - 2$.

Example: Elmhurst College. Gift aid vs. family income (in \$1000s), $n = 50$:

$$\text{estimate} = -0.0431, \quad SE = 0.0108, \quad t_{48}^* = 2.01$$

$$-0.0431 \pm 2.01(0.0108) = (-0.0648, -0.0214)$$

Interpret this interval in context.

Confidence interval for a slope

$$b_1 \pm t_{df}^* \times SE$$

with $df = n - 2$.

Example: Elmhurst College. Gift aid vs. family income (in \$1000s), $n = 50$:

$$\text{estimate} = -0.0431, \quad SE = 0.0108, \quad t_{48}^* = 2.01$$

$$-0.0431 \pm 2.01(0.0108) = (-0.0648, -0.0214)$$

Interpret this interval in context.

We are 95% confident that for each additional \$1,000 of family income, the **average** gift aid falls by between about \$21 and \$65.

Careful wording

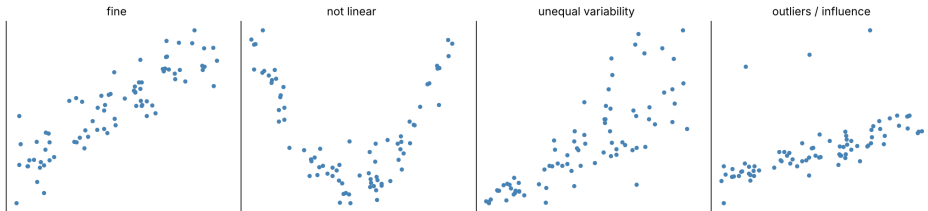
Notice what we did *not* say:

- ▶ ~~“Each \$1,000 more income **causes** aid to drop...”~~ (observational data)
- ▶ ~~“A student with \$1,000 more income gets \$43 less.”~~ (the slope describes an **average** over the population)
- ▶ The interval is for the **slope** β_1 , not for individual students.

Conditions

When can we trust this? LINE

- ▶ **L**inearity: the data should show a linear trend.
- ▶ **I**ndependence: observations are independent of each other.
- ▶ **N**ormality: residuals are nearly normal.
- ▶ **E**qual variability: residuals have about the same spread at every x .



Which conditions matter most?

Which conditions matter most?

- ▶ **Linearity and independence matter most.** If these fail, the slope is estimating the wrong thing, or the SE is wrong.
- ▶ Normality matters mostly for small n and heavy outliers.
- ▶ Equal variability affects the SE; a curved or fanning residual plot is a warning.

Which conditions matter most?

- ▶ **Linearity and independence matter most.** If these fail, the slope is estimating the wrong thing, or the SE is wrong.
- ▶ Normality matters mostly for small n and heavy outliers.
- ▶ Equal variability affects the SE; a curved or fanning residual plot is a warning.

Randomization and bootstrap methods lean less on Normality and Equal variability, so they are a good cross-check when the residual plot looks rough.

Independence in practice: midterm elections

Do unemployment changes predict the president's party's House seat change?

Fitted line (Great Depression years removed): $\hat{y} = -7.36 - 0.89 \times \text{unemp}$

Independence in practice: midterm elections

Do unemployment changes predict the president's party's House seat change?

Fitted line (Great Depression years removed): $\hat{y} = -7.36 - 0.89 \times \text{unemp}$

$$T = \frac{-0.89 - 0}{0.835} \approx -1.07, \quad p = 0.2961$$

Independence in practice: midterm elections

Do unemployment changes predict the president's party's House seat change?

Fitted line (Great Depression years removed): $\hat{y} = -7.36 - 0.89 \times \text{unemp}$

$$T = \frac{-0.89 - 0}{0.835} \approx -1.07, \quad p = 0.2961$$

What do you conclude? What about the independence condition?

Independence in practice: midterm elections

Do unemployment changes predict the president's party's House seat change?

Fitted line (Great Depression years removed): $\hat{y} = -7.36 - 0.89 \times \text{unemp}$

$$T = \frac{-0.89 - 0}{0.835} \approx -1.07, \quad p = 0.2961$$

What do you conclude? What about the independence condition?

Fail to reject: no convincing linear relationship in these data. Elections that are a few years apart are plausibly independent, but time-ordered data are always worth a second look.

Wrap-up

Summary

- ▶ A slope b_1 is a statistic with a sampling distribution.
- ▶ **Test** $H_0 : \beta_1 = 0$ by shuffling the response *or* with $T = b_1/SE$, $df = n - 2$.
- ▶ **Interval** by bootstrapping rows *or* with $b_1 \pm t_{df}^* SE$.
- ▶ Check **L-I-N-E**; linearity and independence first.
- ▶ Tests and intervals agree.

Summary

- ▶ A slope b_1 is a statistic with a sampling distribution.
- ▶ **Test** $H_0 : \beta_1 = 0$ by shuffling the response *or* with $T = b_1/SE$, $df = n - 2$.
- ▶ **Interval** by bootstrapping rows *or* with $b_1 \pm t_{df}^* SE$.
- ▶ Check **L-I-N-E**; linearity and independence first.
- ▶ Tests and intervals agree.

Next: lab, where you run all of this with `lm()`, `broom`, and the `infer` package.